Interactive 5G Evolution Diagram

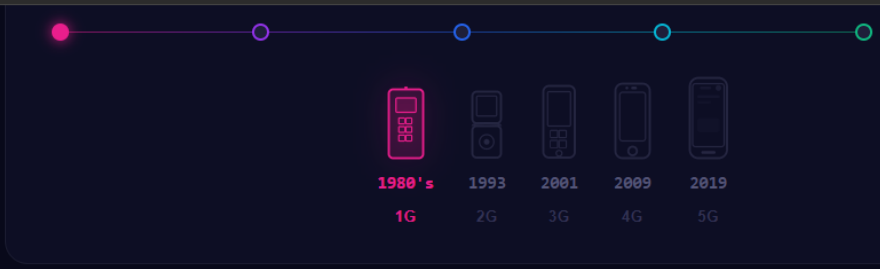
INTERACTIVE TIMELINE

EVOLUTION OF MOBILE COMMUNICATION

From analog voice calls in the 1980s to multi-gigabit 5G — select a generation to explore its technology, speed, and impact.



Interactive 5G Evolution Diagram



1G | 2G | 3G | 4G | 5G

1G 1980S The Analog Beginning

First Generation (1G) launched the mobile era with purely analog voice calls. No encryption, no data — just radio waves carrying voice across towers.

PEAK 0.004 Mbps

KEY SPECS

STANDARD	AMPS (Advanced Mobile Phone System)
FREQUENCY	800 MHz band
PEAK SPEED	~0.004 Mbps (analog voice)
LAUNCHED	1979 Japan / 1983 USA

MILESTONES

- World's first commercial cellular network (NTT, Japan, 1979)
- Calls could be intercepted with a radio scanner
- Handsets weighed ~800 g and cost ~$4,000
- Coverage limited to metropolitan areas only
- No roaming between different carrier networks

Interactive 3G Evolution Diagram

2G

1991

Digital Voice & SMS



Second Generation introduced digital encoding, enabling SMS, MMS, and basic data. GSM became the global standard, transforming mobile from a niche luxury to a mass-market utility.

PEAK **0.384 Mbps** ×96 faster than 1G

KEY SPECS

STANDARDS	GSM, CDMA, TDMA
FREQUENCY	850 / 900 / 1800 / 1900 MHz
PEAK SPEED	0.384 Mbps (EDGE)
LAUNCHED	1991, Finland (Radiolinja)
SECURITY	A5/1 stream cipher (later broken)
KEY FEATURE	SMS text messaging

MILESTONES

- GSM became the world's most widely deployed cellular standard
- First SMS sent on 3 December 1992 (UK)
- GPRS ("2.5G") added packet-switched data in 1997
- EDGE ("2.75G") tripled GPRS throughput in 2003
- Enabled global roaming across 200+ countries

← 1G 3G →

1G 2G 3G 4G 5G

3G

2001

Mobile Internet Arrives



Third Generation delivered broadband-grade speeds to handsets for the first time, making real mobile internet — video calls, streaming, app stores — a daily reality.

PEAK **42 Mbps** ×109 faster than 2G

KEY SPECS

STANDARDS	UMTS / WCDMA, CDMA2000
FREQUENCY	850 / 1700 / 1900 / 2100 MHz
PEAK SPEED	42 Mbps (HSPA+)
LAUNCHED	2001, Japan (NTT DoCoMo)
SECURITY	Mutual authentication (USIM)
KEY FEATURE	Video calling, mobile internet

MILESTONES

- HSPA+ "3.9G" reached 42 Mbps downlink in 2011
- iPhone (2007) made 3G the mainstream internet device
- App Store (2008) created a $1T+ mobile app economy
- First true video calling (3G-324M standard)
- MIMO antenna arrays introduced for capacity gains

← 2G 4G →

Interactive 5G Evolution Diagram

4G — 2009 — All-IP High Speed



Fourth Generation unified mobile on an all-IP architecture, enabling HD streaming, cloud gaming, and always-on connectivity that redefined user expectations worldwide.

PEAK 300 Mbps — ×7 faster than 3G

KEY SPECS

STANDARD	LTE / LTE-Advanced (3GPP Rel. 8–13)
FREQUENCY	600 MHz – 2600 MHz (multi-band)
PEAK SPEED	300 Mbps (Cat.6 LTE-A)
LAUNCHED	2009, Norway & Sweden (TeliaSonera)
LATENCY	~30–50 ms round-trip
KEY FEATURE	Carrier aggregation, 4×4 MIMO

MILESTONES

- LTE-Advanced (Rel.10) met true IMT-Advanced "4G" definition
- VoLTE made voice calls travel entirely over IP
- Carrier aggregation bonded up to 5 spectrum bands
- Enabled HD / 4K video streaming at scale (Netflix, YouTube)
- Massive MIMO and beamforming boosted cell capacity 10×

← 3G — 5G →

5G — 2019 — Ultra-Fast & Ultra-Low Latency



Fifth Generation is not just faster — it's a platform for the connected world: autonomous vehicles, remote surgery, smart cities, and industrial IoT operating at near-real-time responsiveness.

PEAK 20,000 Mbps — ×67 faster than 4G

KEY SPECS

STANDARD	NR (New Radio, 3GPP Rel. 15+)
FREQUENCY	Sub-6 GHz + mmWave (24–100 GHz)
PEAK SPEED	20 Gbps theoretical
LAUNCHED	2019, South Korea (SK Telecom)
LATENCY	<1 ms (URLLC slices)
KEY FEATURE	Network slicing, massive IoT

MILESTONES

- mmWave frequencies unlock multi-Gbps throughput in dense areas
- Network slicing creates virtual dedicated networks per use-case
- Supports 1 million devices per km² (vs 4G's 100,000)
- Enables tactile internet: haptic feedback over wireless
- SA (Standalone) core eliminates reliance on 4G infrastructure

← 4G — End of timeline